

NETWORKING

A network is a collection of devices connected together so they can communicate and share resources.

Example:



Examples of devices in a network:

- Laptop
- Mobile
- Printer
- Router
- Switch
- Server

OSI Model

The OSI model explains how data travels from one computer to another.

7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical

Layer 7 – Application

User-facing protocols.

Examples:

- HTTP
- HTTPS
- FTP
- SMTP

- DNS

Example:

You open:

www.google.com

The browser creates an HTTP request.

Layer 6 – Presentation

Responsible for:

- Encryption
- Compression
- Data formatting

Examples:

- SSL/TLS
- JPEG
- PNG
- MP4

HTTPS uses TLS encryption here.

Layer 5 – Session

Manages communication sessions.

Example:

Zoom Meeting

The session stays active until you disconnect.

Layer 4 – Transport

Protocols:

- TCP
- UDP

Responsibilities:

- Reliability
 - Error checking
 - Flow control
 - Segmentation and reassembly
-

Layer 3 – Network

Responsible for routing packets using **IP addresses**.

Devices:

- Router

Protocols:

- IPv4
 - IPv6
 - ICMP
-

Layer 2 – Data Link

Uses **MAC addresses**.

Device:

- Switch

Responsible for communication within the same local network.

Layer 1 – Physical

Includes the actual hardware.

Examples:

- Ethernet cables
- Fibre optic cables
- Wireless signals

What is a Protocol?

A **protocol** is a set of rules that defines how devices communicate over a network.

Example:

When you speak English with another person, both of you understand the same language. Similarly, computers use protocols to communicate.

Computer A ----HTTP----> Computer B

Without protocols, computers would not understand each other.

Application Layer Protocols

These protocols are used directly by applications like web browsers, email clients, and file transfer tools.

Protocol	Full Form	Default Port	Purpose
HTTP	HyperText Transfer Protocol	80	Transfers web pages over the Internet.
HTTPS	HyperText Transfer Protocol Secure	443	Secure version of HTTP using SSL/TLS encryption.
FTP	File Transfer Protocol	20, 21	Transfers files between computers.
SFTP	SSH File Transfer Protocol	22	Secure file transfer over SSH.
TFTP	Trivial File Transfer Protocol	69	Lightweight file transfer, often used for network device booting.
SMTP	Simple Mail Transfer Protocol	25, 587, 465	Sends emails from clients or mail servers.
POP3	Post Office Protocol Version 3	110	Downloads emails from a mail server to a client.
IMAP	Internet Message Access Protocol	143	Accesses and synchronises emails while keeping them on the server.
DNS	Domain Name System	53	Converts domain names into IP addresses.

Protocol	Full Form	Default Port	Purpose
DHCP	Dynamic Host Configuration Protocol	67, 68	Automatically assigns IP addresses to devices.
SNMP	Simple Network Management Protocol	161, 162	Monitors and manages network devices.
NTP	Network Time Protocol	123	Synchronises the system time of computers.
LDAP	Lightweight Directory Access Protocol	389	Accesses directory services such as Active Directory.
LDAPS	LDAP Secure	636	Secure version of LDAP.
Telnet	Telecommunication Network	23	Remote login without encryption (rarely used today).
SSH	Secure Shell	22	Secure remote login and command execution.
RDP	Remote Desktop Protocol	3389	Provides a graphical remote desktop connection.
SIP	Session Initiation Protocol	5060	Establishes VoIP calls.
RTP	Real-time Transport Protocol	Dynamic	Carries voice and video streams.
RTSP	Real-Time Streaming Protocol	554	Controls streaming media sessions.
MQTT	Message Queuing Telemetry Transport	1883, 8883	Lightweight messaging protocol used in IoT.
AMQP	Advanced Message Queuing Protocol	5672	Reliable messaging between applications.

Transport Layer Protocols

These protocols handle end-to-end communication.

TCP (Transmission Control Protocol)

Port: Uses many application ports

Purpose

- Reliable communication
- Ensures packets arrive correctly
- Retransmits lost packets
- Maintains packet order

Used by

- HTTP
 - HTTPS
 - SSH
 - FTP
 - SMTP
 - MySQL
-

UDP (User Datagram Protocol)

Purpose

- Faster than TCP
- No guarantee of delivery
- No retransmission
- Lower overhead

Used by

- DNS
 - VoIP
 - Video streaming
 - Online gaming
-

Internet Layer Protocols

IP (Internet Protocol)

Purpose:

Assigns addresses to devices and routes packets between networks.

Versions:

- IPv4
- IPv6

Example:

192.168.1.100

ICMP (Internet Control Message Protocol)

Purpose:

Used for diagnostics and error reporting.

Example:

ping google.com

Uses ICMP Echo Request and Echo Reply messages.

ARP (Address Resolution Protocol)

Purpose:

Finds the MAC address corresponding to an IPv4 address on the local network.

Example:

IP Address

↓

ARP

↓

MAC Address

RARP (Reverse Address Resolution Protocol)

Purpose:

Maps a MAC address to an IP address.

Largely obsolete and replaced by newer protocols such as BOOTP and DHCP.

IGMP (Internet Group Management Protocol)

Purpose:

Manages multicast group membership.

Commonly used for IPTV and streaming.

Routing Protocols

These protocols help routers exchange routing information.

Protocol	Full Form	Purpose
RIP	Routing Information Protocol	Simple distance-vector routing.
OSPF	Open Shortest Path First	Link-state routing used in enterprise networks.
EIGRP	Enhanced Interior Gateway Routing Protocol	Cisco routing protocol.
BGP	Border Gateway Protocol	Routes traffic between organisations and Internet providers.

Security Protocols

SSL (Secure Sockets Layer)

Older protocol used to encrypt communication.

Now considered obsolete.

TLS (Transport Layer Security)

Modern replacement for SSL.

Used by:

- HTTPS
 - Secure email
 - APIs
-

IPSec (Internet Protocol Security)

Provides secure IP communication using encryption and authentication.

Often used in VPNs.

Kerberos

Authentication protocol used in Windows Active Directory.

VPN Protocols

Protocol	Purpose
PPTP	Older VPN protocol (not recommended for new deployments).
L2TP	Layer 2 Tunnelling Protocol, often combined with IPSec.
IPSec	Secure VPN communication.
OpenVPN	Popular open-source VPN protocol.
WireGuard	Modern, lightweight VPN protocol.

File Sharing Protocols

Protocol	Purpose
SMB	File and printer sharing in Windows networks.
CIFS	Older implementation of SMB.
NFS	Network File System for Linux/Unix systems.

Email Protocols

Protocol	Purpose
SMTP	Sends email.
POP3	Downloads email to a client.
IMAP	Synchronises email while keeping it on the server.

DevOps-Specific Protocols

Protocol	Used In
SSH	Linux servers, Git, Ansible, Jenkins
HTTPS	GitHub, GitLab, Jenkins, Kubernetes APIs
DNS	Kubernetes Services, AWS Route 53
NTP	Time synchronisation across servers
TLS	Kubernetes, Docker Registry, Ingress
HTTP	Web applications and REST APIs
MQTT	IoT deployments
AMQP	RabbitMQ messaging
gRPC	High-performance microservice communication using HTTP/2 and Protocol Buffers
WebSocket	Real-time, full-duplex communication for dashboards and web apps

Common Protocols and Default Ports

Protocol	Port
FTP	20, 21
SSH	22
Telnet	23
SMTP	25
DNS	53

Protocol	Port
DHCP Server	67
DHCP Client	68
TFTP	69
HTTP	80
POP3	110
NTP	123
IMAP	143
SNMP	161, 162
LDAP	389
HTTPS	443
SMB	445
SMTPS	465
RTSP	554
Submission (SMTP)	587
LDAPS	636
IMAPS	993
POP3S	995
MySQL	3306
PostgreSQL	5432
RDP	3389
Kubernetes API Server	6443
Redis	6379
SonarQube	9000
Prometheus	9090
Grafana (default)	3000
Jenkins (default)	8080

Protocol	Port
RabbitMQ	5672
MQTT	1883

Protocols Every DevOps Engineer Should Master

Focus on these first, as they appear frequently in interviews and daily work:

1. TCP
2. UDP
3. IP (IPv4/IPv6)
4. ARP
5. ICMP
6. DNS
7. DHCP
8. HTTP
9. HTTPS
10. SSH
11. FTP
12. SFTP
13. SMTP
14. POP3
15. IMAP
16. TLS
17. NTP
18. SNMP
19. LDAP
20. SMB/NFS

Linux Networking Commands Cheat Sheet

Command	Syntax	Meaning	Purpose	Example
ip addr	ip addr	Displays all network interfaces	Check IP configuration	ip addr

Command	Syntax	Meaning	Purpose	Example
		with IP addresses		
ip a	ip a	Short form of ip addr	View IP addresses	ip a
ip -4 addr	ip -4 addr	Shows only IPv4 addresses	Check IPv4	ip -4 addr
ip -6 addr	ip -6 addr	Shows only IPv6 addresses	Check IPv6	ip -6 addr
ip link	ip link	Shows network interfaces and their status	View interface status	ip link
ip link set eth0 up	sudo ip link set eth0 up	Enables a network interface	Bring interface online	sudo ip link set eth0 up
ip link set eth0 down	sudo ip link set eth0 down	Disables a network interface	Bring interface offline	sudo ip link set eth0 down
ip route	ip route	Displays routing table	View routes	ip route
ip neigh	ip neigh	Displays ARP cache (IP ↔ MAC mapping)	Check neighbour devices	ip neigh
ifconfig	ifconfig	Displays interface details (older command)	View network configuration	ifconfig
route -n	route -n	Displays routing table (older command)	View routes	route -n

Command	Syntax	Meaning	Purpose	Example
dig +short	dig +short google.com	Shows only IP address	Quick DNS lookup	dig +short google.com
host	host google.com	Performs DNS lookup	Check DNS	host google.com
netstat -tuln	netstat -tuln	Shows listening TCP/UDP ports	Check open ports	netstat -tuln
netstat -an	netstat -an	Shows all network connections	Monitor connections	netstat -an
ss -tulpn	ss -tulpn	Shows listening ports with process names	Check services	ss -tulpn
ss -t	ss -t	Shows TCP connections	View TCP sessions	ss -t
ss -u	ss -u	Shows UDP connections	View UDP sessions	ss -u
curl	curl https://example.com	Fetches web content	Test website/API	curl https://google.com
curl -I	curl -I https://google.com	Shows HTTP response headers	Test web server	curl -I https://google.com
wget	wget https://example.com/file.zip	Downloads files	Retrieve files	wget https://example.com/file.zip
ssh	ssh user@192.168.1.10	Secure remote login	Connect to remote server	ssh ec2-user@54.x.x.x
scp	scp file.txt user@host:/home/user	Securely copies files	Transfer files	scp test.txt user@server:/tmp
sftp	sftp user@host	Secure file transfer session	Upload/download files	sftp ec2-user@server

Command	Syntax	Meaning	Purpose	Example
nc	nc -zv google.com 443	Tests whether a port is open	Check service availability	nc -zv localhost 8080
telnet	telnet google.com 80	Connects to a remote port	Test connectivity	telnet localhost 8080
lsof -i	lsof -i :8080	Shows which process uses a port	Find process by port	sudo lsof -i :8080
nmap	nmap 192.168.1.10	Scans ports on a host	Security testing	nmap 192.168.1.10
tcpdump	tcpdump -i eth0	Captures network packets	Analyse traffic	sudo tcpdump -i eth0
ethtool	ethtool eth0	Displays NIC information	Check link speed	sudo ethtool eth0
nmcli device status	nmcli device status	Shows network device status	NetworkManager status	nmcli device status
iptables -L	iptables -L	Lists firewall rules	Check firewall	sudo iptables -L
firewall-cmd --list-all	firewall-cmd --list-all	Displays firewall configuration	Check allowed services	sudo firewall-cmd --list-all
resolvectl status	resolvectl status	Shows DNS resolver configuration	Verify DNS settings	resolvectl status
ipcalc	ipcalc 192.168.1.0/24	Calculates subnet details	Subnet calculations	ipcalc 10.0.0.0/16
whois	whois google.com	Displays domain	Domain lookup	whois example.com

Command	Syntax	Meaning	Purpose	Example
		registration information		

★ Most Important Commands for DevOps Interviews

Command	Interview Purpose
ip addr	Find the server IP address
ip route	Check routing configuration
ping	Verify network connectivity
hostname -I	Display local IP
nslookup / dig	Troubleshoot DNS
ss -tulpn	Check running services and listening ports
curl	Test web applications and REST APIs
ssh	Connect to Linux servers
scp	Copy files securely
lsof -i :PORT	Identify which process is using a port
nmap	Scan ports on a server
tcpdump	Capture and analyse network traffic
iptables / firewall-cmd	Verify firewall rules
ethtool	Check network interface health